

Miguel Romero

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EDUCATION

Pennsylvania State University

Bachelor of Science in Computer Science

University Park, PA

Aug. 2022 – May 2026

- **Relevant Coursework:** Database Management Systems, Machine Learning and Algorithmic AI, Artificial Intelligence, Exploratory Data Mining, Intro to Cryptography

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, C, C++, Java, Verilog

ML & AI: PyTorch, NumPy, Pandas, scikit-learn, Ollama, LM Studio

Backend & Data: FastAPI, SQLAlchemy, Pydantic, SQLite, Power BI, MS Project, REST APIs

Cloud & Tools: AWS, Azure AI, Docker, Git, n8n, VS Code, PyCharm, IntelliJ

PROJECTS

Plant Operations Automation | *Python, SQLite, n8n, Power BI, Webhooks, Local LLM*

2026

- Modeled a 24/7 plastic-film facility in a 16-table SQLite schema covering 293 orders, 809 logged rolls, and 1,800 ledger rows across a six-week production horizon.
- Built four trigger pipelines on the shared schema (cron briefing, threshold material forecast, calendar webhook, and a human approval gate), with a rejection branch that returns invalid schedule changes with a reason instead of corrupting state.
- Kept scheduling decisions deterministic and constraint-checked for auditability, using an LLM only to narrate finished decisions into plain-language briefings.

Inbox Orchestration | *n8n, Outlook, LM Studio, Qwen2.5-7B, Excel, Discord*

2026

- Built a multi-agent pipeline that routes every incoming email through a local 7B orchestrator, which classifies each message and drops marketing and noise before invoking a specialist agent.
- Designed an approval gate so drafted replies post to Discord and wait for a human decision before anything is sent, and a second agent that extracts company, role, status, and date from job-related email and appends structured rows to a tracking sheet.
- Ran the system against 1,000 real messages, logging 851 appended rows and three concrete failure modes (duplicate rows, hallucinated fields, unnecessary tool calls) to scope the next fix.

Chess Coach AI | *Python, FastAPI, SQLAlchemy, Pydantic, Ollama, React*

2026

- Built an NLP coaching pipeline that ingests games from the chess.com API, parses PGN, scores every ply with Stockfish, and aggregates centipawn loss by game phase into recurring mistake patterns.
- Grounded LLM output in a Pydantic schema the model must populate, constraining generated explanations to engine data instead of generic advice.
- Implemented LLM tool calling against a live board sandbox so the model can replay candidate lines before explaining them; runs fully local via Ollama, so no game data leaves the machine.

Multi-Agent UAV Path Planning | *PyTorch, MAPPO, NumPy, NVIDIA Isaac Sim*

2026

- Trained an 8-agent coordination policy using MAPPO with a centralized critic and a parameter-shared Gaussian actor, applying centralized training with decentralized execution so each agent acts on local observations alone.
- Combined global A* routing with reciprocal velocity obstacle avoidance, and built an independent collision auditor that scores every frame without steering, making violations measurable failures rather than tuning artifacts.

EXPERIENCE

Packer

Multi-Plastics Extrusion

Hazleton, PA

May 2025 – Sep. 2025

- Packed and staged finished film rolls on 12-hour night shifts, inspecting output for defects and flagging quality issues under continuous throughput targets.

Resident Assistant

Penn State University

Hazleton, PA

Aug. 2023 – May 2024

- Supported 40+ first-year residents through conflict resolution, mentorship, and community events; coordinated with housing staff on policy enforcement and campus resource referrals.